IMO 2026 — Problem 5

Solution by GPT-5.6 Sol (lightweight, xhigh reasoning)

2026-07-18

Source: `results/imo-2026/gpt-5.6-sol-lightweight-xhigh/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

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Approaches tried

- **Direct equality and orbit analysis (successful):** Substituting $x = f(y)$ makes the two outer expressions equal, hence forces $f(f(y)) = 2f(y) - y$. Forward iteration then shows that $f(y) - y$ is nonnegative and remains constant along the whole forward orbit.
- **Asymptotic comparison of two orbits (successful):** If two points have positive displacements $a = f(u) - u$ and $b = f(v) - v$, their forward orbits are arithmetic progressions. Choosing far-out terms of the two progressions that remain a bounded distance apart makes the RMS–AM gap tend to zero, forcing $a = b$.
- **Topological elimination of mixed behavior (successful):** Once every displacement is either 0 or one common $c > 0$, the first inequality shows that the zero-displacement and c-displacement sets are both open. Connectedness of $\mathbb{R}_{>0}$ rules out a mixture.
- **Ratio/asymptotic-bound exploration (abandoned):** Writing $f(t) = t\,r(t)$ gives pairwise quadratic constraints and rough bounds, but these do not by themselves expose the exact solutions as efficiently as the equality substitution $x = f(y)$.

Current best

All solutions are

$$\boxed{f(x) = x + c \quad \text{for every } x > 0, \qquad c \geq 0.}$$

The proof below derives an arithmetic-progression structure for every forward orbit, proves that all positive orbit increments coincide, and then excludes coexistence of zero and positive increments.

Full proof

Put

$$d(t) = f(t) - t \qquad (t > 0).$$

We first determine the forward orbits of f. In the given inequalities, fix $y > 0$ and set $x = f(y)$. Both outer expressions then equal $f(y)$:

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \quad \text{and} \quad \sqrt{f(y)f(y)} = f(y).$$

Consequently the middle expression must also equal $f(y)$, so

$$\frac{f(f(y)) + y}{2} = f(y), \qquad \text{or equivalently} \qquad f(f(y)) = 2f(y) - y. \qquad (1)$$

For any fixed $t > 0$, define $t_0 = t$ and $t_{n+1} = f(t_n)$ for $n \geq 0$. Applying (1) to t_n gives

$$t_{n+2} = 2t_{n+1} - t_n.$$

Thus, by induction,

$$t_n = t + n(f(t) - t) = t + nd(t) \qquad (n \geq 0). \qquad (2)$$

Every t_n belongs to $\mathbb{R}_{>0}$. If $d(t) < 0$, the right-hand side of (2) is nonpositive for all sufficiently large integers n, a contradiction. Hence

$$d(t) \geq 0 \qquad \text{for every } t > 0. \qquad (3)$$

Moreover, (2) shows that along this orbit

$$f(t + nd(t)) = t + (n+1)d(t), \qquad (4)$$

so its displacement remains equal to $d(t)$.

We next prove that any two positive displacements are equal. Let $u, v > 0$, and write

$$a = d(u) > 0, \qquad b = d(v) > 0.$$

Suppose, without loss of generality, that $a > b$. By (4), for all nonnegative integers n, m,

$$f(u + na) = u + (n+1)a, \qquad f(v + mb) = v + (m+1)b. \qquad (5)$$

Choose integers $n \to \infty$ and

$$m_n = \left\lfloor \frac{u + na - v - b}{b} \right\rfloor.$$

For all sufficiently large n, $m_n \geq 0$. Set

$$p_n = u + na, \qquad q_n = v + (m_n + 1)b.$$

The definition of m_n gives

$$0 \leq p_n - q_n < b, \qquad (6)$$

and both p_n and q_n tend to infinity.

Apply the first given inequality with $x = p_n$ and $y = v + m_n b$. By (5), $f(x) = p_n + a$ and $f(y) = q_n$, and hence

$$p_n + q_n + (a - b) = f(x) + y \leq \sqrt{2(p_n^2 + q_n^2)}.$$

It follows that

$$0 < a - b \leq \sqrt{2(p_n^2 + q_n^2)} - p_n - q_n = \frac{(p_n - q_n)^2}{\sqrt{2(p_n^2 + q_n^2)} + p_n + q_n}. \qquad (7)$$

By (6), the numerator on the right is less than b^2, whereas its denominator tends to infinity. The right-hand side of (7) therefore tends to 0, contradicting $a - b > 0$. Interchanging u and v similarly rules out $b > a$. Thus

$$d(u) = d(v) \quad \text{whenever } d(u) > 0 \text{ and } d(v) > 0. \qquad (8)$$

If d is identically zero, then $f(t) = t$, which is one of the claimed functions. Otherwise, by (8), there is a constant $c > 0$ such that

$$d(t) \in \{0, c\} \qquad \text{for every } t > 0. \qquad (9)$$

Define

$$A = \{t > 0 : d(t) = 0\}, \qquad B = \{t > 0 : d(t) = c\}.$$

We show that both A and B are open in $\mathbb{R}_{>0}$. If this were false at some point of either set, there would be sequences $r_k \in B$ and $s_k \in A$ with $r_k - s_k \to 0$ and with both sequences converging to the same positive limit L (one can keep whichever of the two points is fixed and choose points of the other set approaching it). Apply the first inequality with $x = r_k$ and $y = s_k$. Since $f(r_k) = r_k + c$ and $f(s_k) = s_k$, it yields

$$r_k + s_k + c = f(r_k) + s_k \leq \sqrt{2(r_k^2 + s_k^2)}. \qquad (10)$$

Taking $k \to \infty$ in (10) gives $2L + c \leq 2L$, impossible. Therefore every point of either A or B has a neighborhood containing no point of the other set; both sets are open.

The disjoint open sets A and B have union $\mathbb{R}_{>0}$. Since $\mathbb{R}_{>0}$ is connected and B is nonempty, we must have $A = \emptyset$. Hence $d(t) = c$ for every $t > 0$. Together with the identically-zero case, this proves that every possible function has the form

$$f(t) = t + c$$

for some constant $c \geq 0$.

Finally, every such function does satisfy the inequalities. Indeed, if $f(t) = t + c$ with $c \geq 0$, then for all $x, y > 0$,

$$\sqrt{\frac{x^2 + f(y)^2}{2}} = \sqrt{\frac{x^2 + (y+c)^2}{2}} \geq \frac{x + (y+c)}{2} = \frac{f(x) + y}{2} \geq \sqrt{x(y+c)} = \sqrt{xf(y)},$$

where the two inequalities are respectively the root-mean-square–arithmetic-mean and arithmetic-mean–geometric-mean inequalities for the positive numbers x and $y + c$. This completes the characterization.